Knowledge Graph Project - Complete Explanation

Project Objective

The goal of this project is to transform unstructured text documents into a structured Knowledge Graph (KG). Instead of keeping information as plain sentences, we extract entities and relationships so that computers can reason over them.

Overall Pipeline

20 Newsgroups Dataset → Load Dataset → Text Cleaning → Named Entity Recognition (NER) → Relation Extraction → Subject-Relation-Object Triples → Knowledge Graph Construction → Visualization → Graph Analysis → Graph Export.

Step 1: Dataset Loading

We use the built-in 20 Newsgroups dataset from scikit-learn containing about 11,314 documents across 20 categories. The data is loaded into a Pandas DataFrame for easier processing and analysis.

Step 2: Text Cleaning

The raw text contains URLs, email addresses, punctuation, numbers, headers, and unnecessary whitespace. Cleaning removes this noise so that NLP models can extract more accurate entities and relationships.

Step 3: Named Entity Recognition (NER)

spaCy identifies important real-world entities such as PERSON, ORGANIZATION, LOCATION, PRODUCT, DATE, and EVENT. Every extracted entity becomes a node in the Knowledge Graph.

Step 4: Relation Extraction

Using spaCy dependency parsing, each sentence is analyzed to identify the grammatical subject, verb, and object. These are converted into Subject-Relation-Object triples such as (Microsoft, acquired, GitHub).

Step 5: Triple Store

All extracted triples are stored in a structured table. Triples are the fundamental representation used by modern Knowledge Graphs.

Step 6: Knowledge Graph Construction

NetworkX MultiDiGraph is used to create the graph. Subjects and objects become nodes while relations become directed edges. A MultiDiGraph allows multiple relationships between the same pair of entities.

Step 7: Visualization

The graph is visualized using NetworkX and Matplotlib to help users understand entity connections and debug the extraction process.

Step 8: Graph Export

The graph is exported as GraphML and GEXF so it can be opened in Neo4j, Gephi, Cytoscape, and other graph analysis tools.

Step 9: Graph Analysis

Graph statistics such as node count, edge count, degree centrality, PageRank, connected components, and communities are computed to understand the structure and importance of entities.

Why This Is a Knowledge Graph

The project represents knowledge as entities connected through meaningful semantic relationships stored as Subject-Relation-Object triples.

Limitations

The project uses dependency-based relation extraction, which may miss complex relationships, does not resolve pronouns, does not perform entity linking, and does not use a formal ontology.

Future Improvements

Potential extensions include transformer-based NER, REBEL or LLM-based relation extraction, Wikidata entity linking, Neo4j integration, RDF stores, Node2Vec embeddings, Graph Neural Networks (GNNs), GraphRAG, and multimodal knowledge graphs.

Final Workflow

Raw Text → Cleaning → NER → Relation Extraction → Triples → Knowledge Graph → Visualization → Analysis → Export → Foundation for GraphRAG, Neo4j, and GNN applications.